Is the model accurate?

- **Every test score is a claim:** this is how the model will do on data it has not seen
- **The claim is easy to get wrong on climate data,** and the mistake makes the model look better than it is
- **Four main steps:** establish a baseline, use a split that matches the application, chose settings on validation set (or use cross-validation on the training set), and test the final model on the test set

A baseline

climatology, persistence, or the mean. No score means anything on its own

↓

A split that matches the application

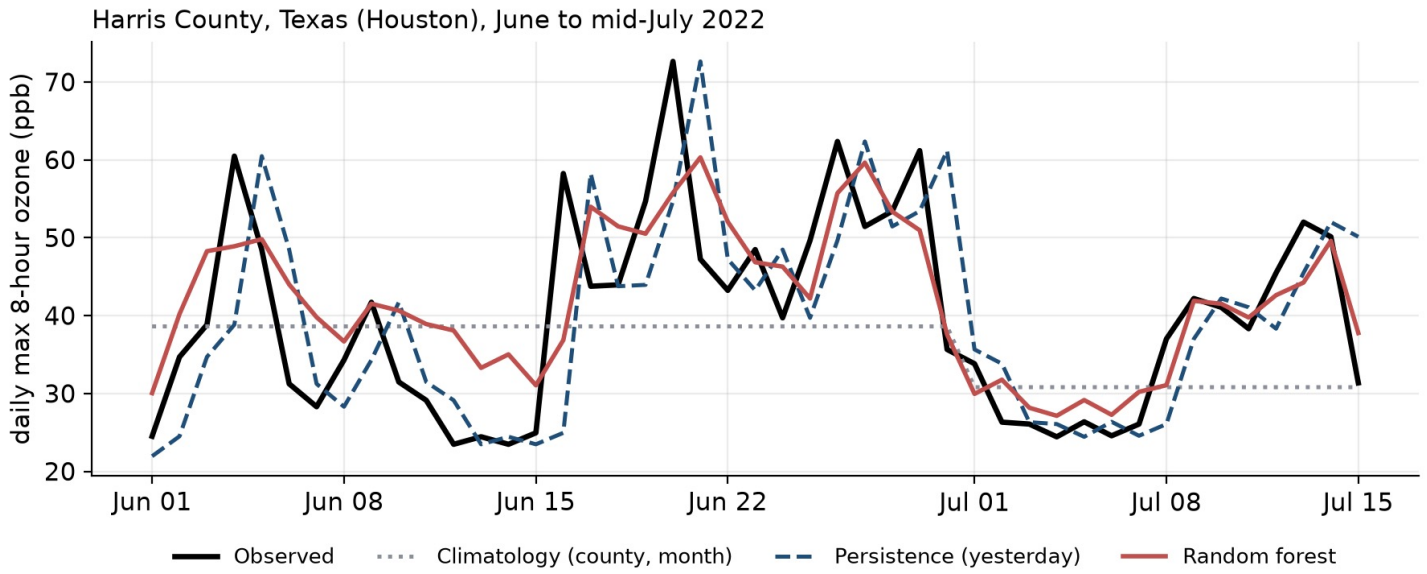
many splits, not one, and folds that respect space and time

↓

Hyperparameters (model settings) chosen with cross-validation

validation data or cross-validation chooses them; the test set is scored once

Always compare against a baseline

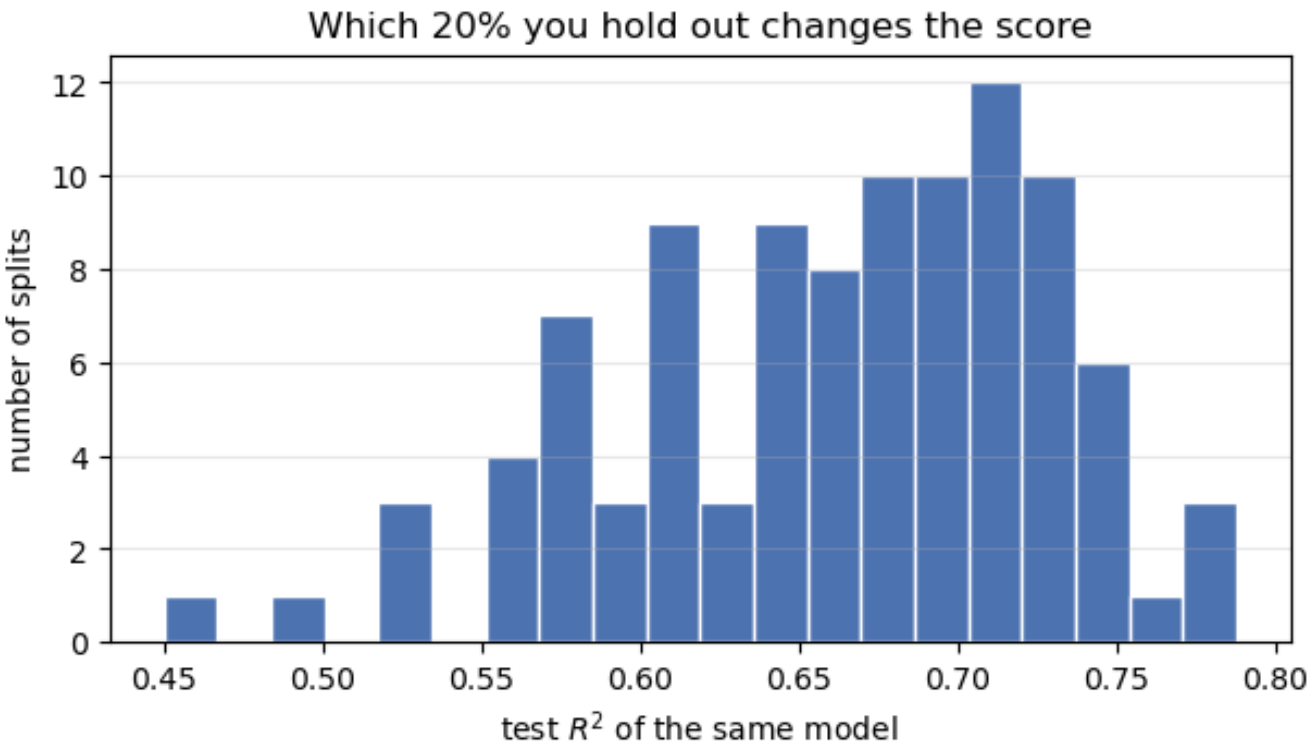


2022, 47,300 county-days	RMSE (ppb)	R^2
Climatology	8.4	0.44
Persistence	7.9	0.51
Random forest	5.9	0.73

- **Climatology:** the mean for that county and month, from the training years
- **Persistence:** yesterday's value. Hard to beat, because weather carries over from day to day
- **The forest beats both:** a quarter less error than persistence, from yesterday's ozone plus today's weather. That gap is the reason to build the model
- **A skill score without its baseline** is not a result

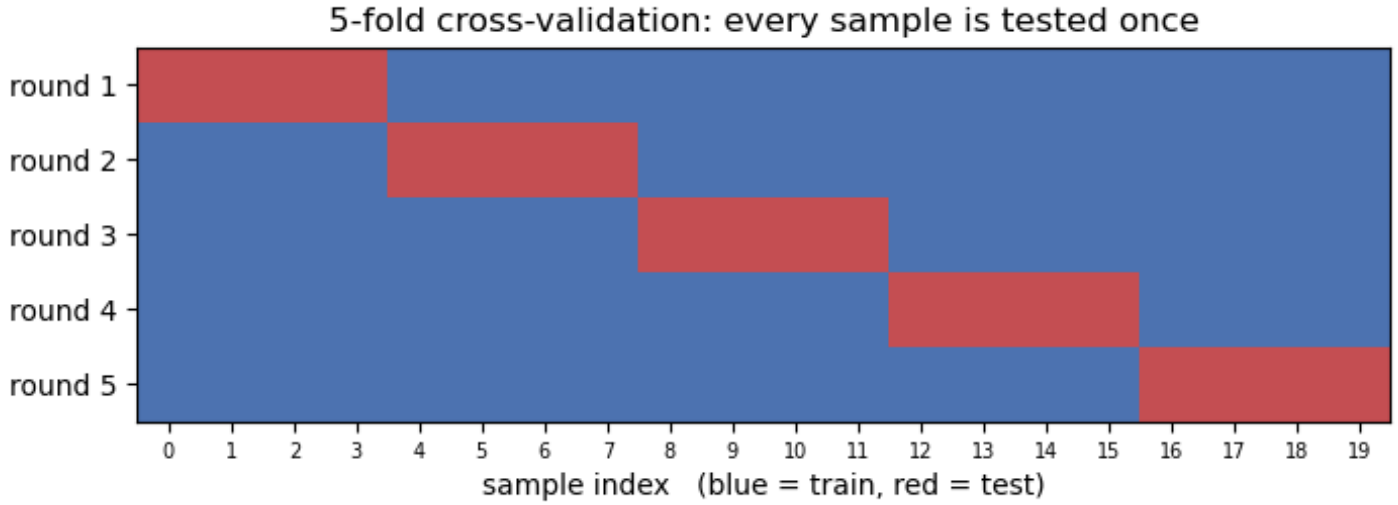
EPA Air Quality System daily maximum 8-hour ozone, county means, 172 counties in 50 states, 2019 to 2023. Forest trained on 2019 to 2021 with today's maximum temperature, wind, humidity, day of year, location, and yesterday's ozone

One hold-out score is one noisy number



- **Same model, same 300 samples, 100 random 80/20 splits**
- **Test R^2 runs from 0.45 to 0.79.** Which 60 samples you held out decided the score
- **A single hold-out is one draw from this histogram,** and you do not know where in it you landed
- **Small data sets make it worse,** and station records, field campaigns, and model ensembles are small

K-fold cross-validation

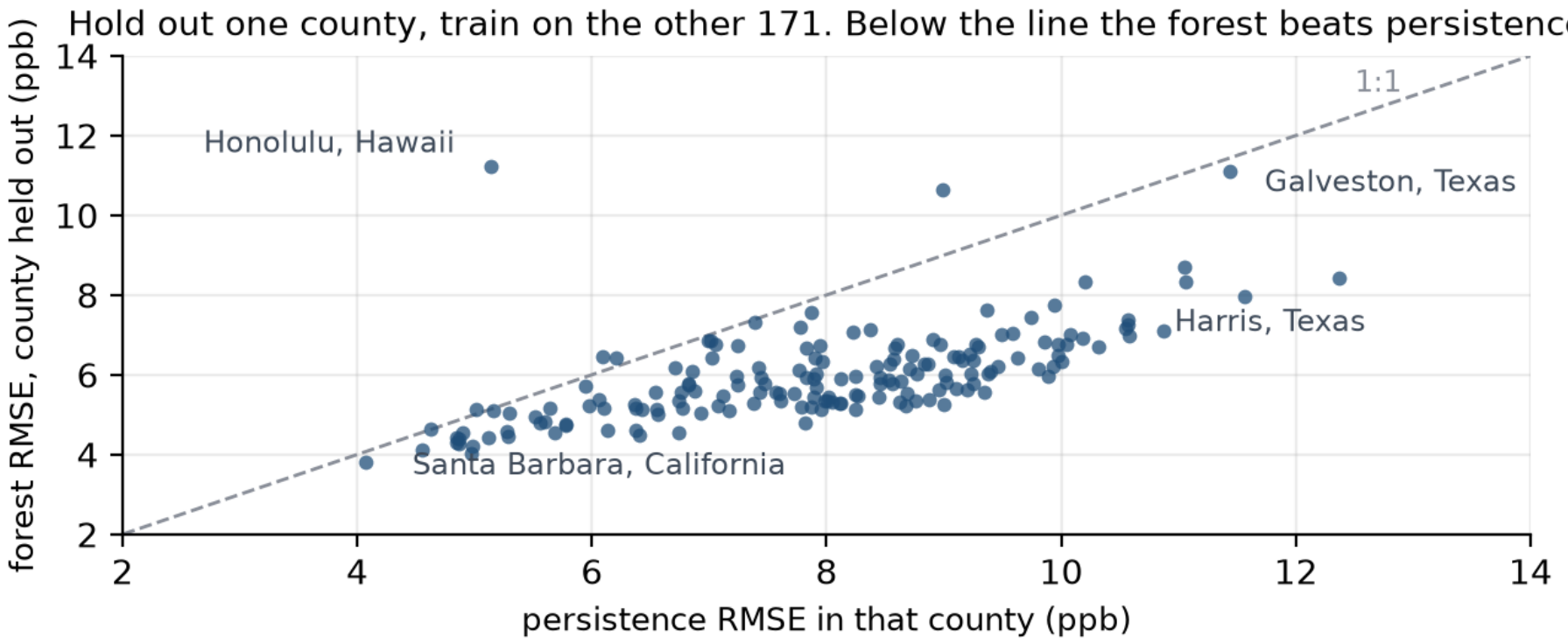


- **Cut the data into k folds.** Hold out each in turn, fit on the rest, score on the held-out fold
- **Every sample is tested once** and trained on k − 1 times
- **Report the mean and the spread** of the k scores, not one number
- **k = 5 or 10 is standard.**

What it buys: all the data tested, most of the data trained on in every round, and a spread.
What it costs: k fits instead of one.

- **Cross validation i**s a way to estimate how well a model will do on data it has not seen, by repeatedly holding out part of the training data, fitting on the rest, and scoring on the held-out part. The scores from the repeats are averaged (and their spread reported) to give an estimate of generalization error.
  - For cross validation, the folds already play the role of the validation dataset, so no need to split twice.
  - A separate validation set is sometimes still needed, especially with neural networks, or if fitting the model “k” times is computationally expensive

Leave-one-out: hold out one site, predict it



- **Leave one group out:** train on 171 counties, predict the one the model has never seen, 172 times. The question a new monitoring site asks
- **Each round scores one county.** 163 of 169 beat persistence. Pooled, 6.02 ppb, the same answer as GroupKFold's 6.07, from 172 fits instead of five
- **The failures are the lesson.** Honolulu: persistence 5.1 ppb, the forest 11.2. Trained on the mainland, it has never seen a tropical island. A site outside the training set is extrapolation

The assumption underneath: independent samples

- **Random k-fold assumes samples are independent and identically distributed:** knowing one tells you nothing about another
- **Geophysical data violate this almost everywhere**
- **A random split then puts near-copies of each test sample into the training set.**
- **The error is silent.** Nothing in the code warns you

In space

adjacent grid cells and nearby stations are near-duplicates

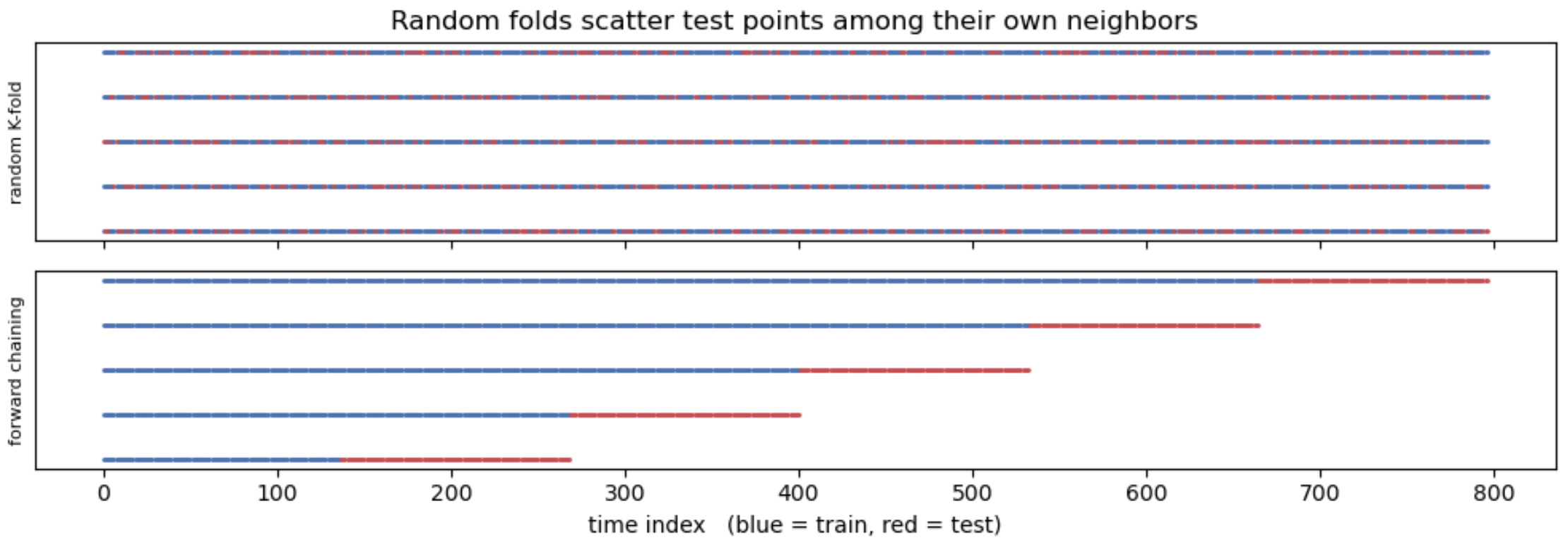
In time

today looks like yesterday; a year of daily rows is far fewer than 365 independent samples

Across the sample

one satellite orbit, one instrument, one model ensemble: shared structure

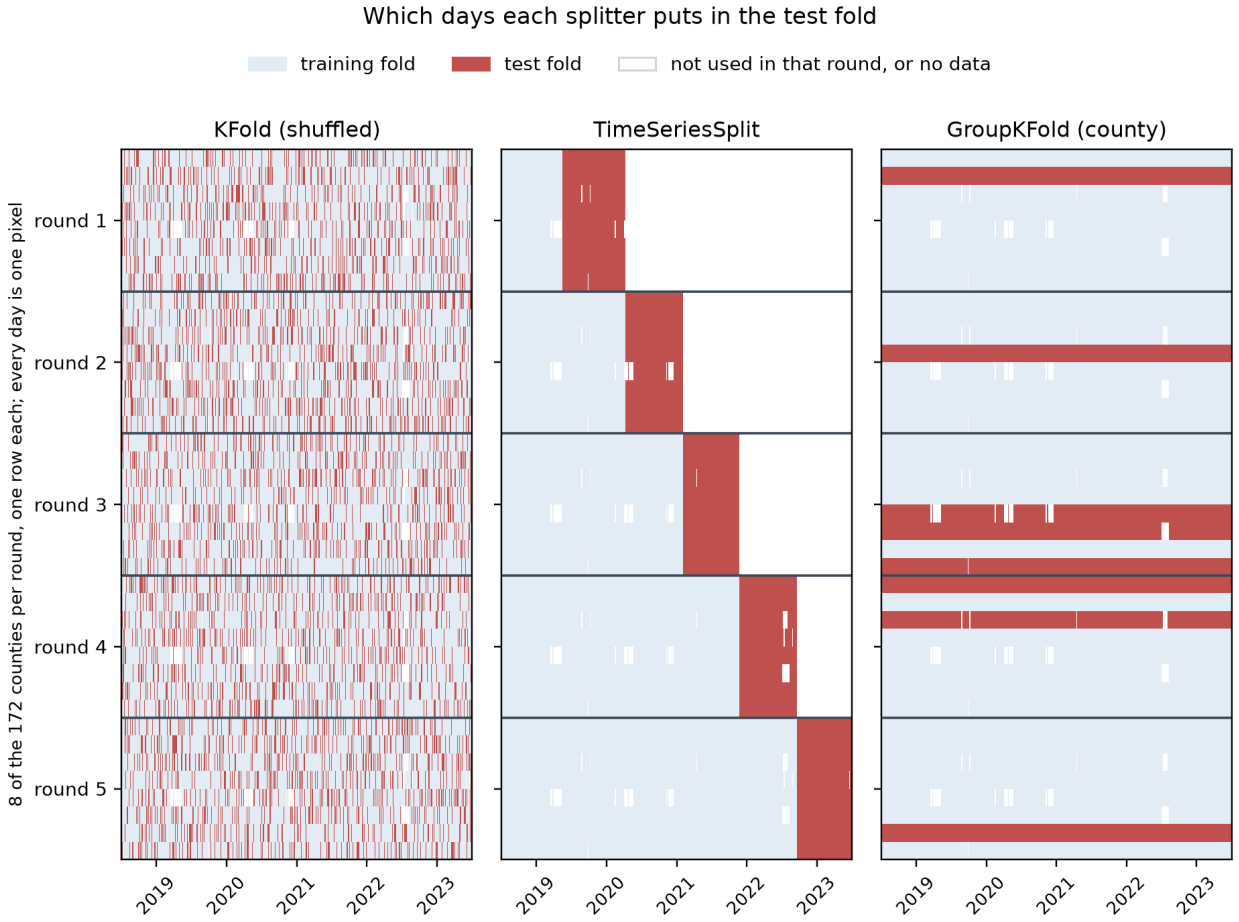
Time: random folds train on the future



Same forest, same data	R^2
Random 5-fold	0.44
Forward chaining (TimeSeriesSplit)	−0.36
Persistence	0.94

- **A slow unobserved mode makes the target unlearnable from the features**
- **Random folds say 0.44:** a model that seems to work
- **Forward chaining says worse than the mean.** That is the truth, and persistence beats everything
- **The random score was not inflated, it was invented,** from the model recognizing its neighbors in time

Three splitters, three questions



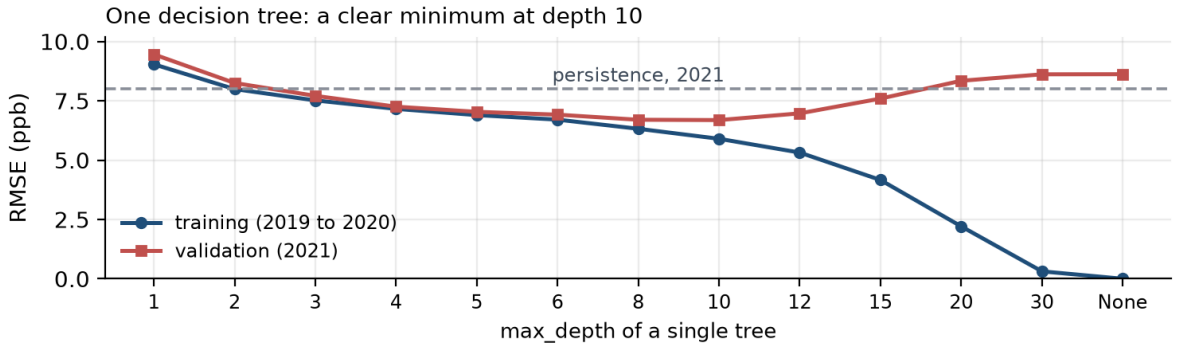
Split, 100-tree forest	RMSE (ppb)	Persistence
A: KFold, shuffled	5.67 ± 0.02	7.97
B: TimeSeriesSplit	6.03 ± 0.22	7.97
C: GroupKFold by county	6.07 ± 0.17	7.97
Test year 2022, trained on 2019 to 2021	5.86	7.95

- **Each splitter asks a different question:** fill in a missing day, forecast next year, or handle a county with no monitor
- **The random split flatters by 0.4 ppb.** It tests each day with its neighbors in the training set. The time and county splits agree with each other and with the test year
- **Same data, same code, same forest.** The split is the question. This model passes all three, which is what makes it deployable

EPA Air Quality System daily maximum 8-hour ozone, county means, 172 counties in 50 states, 2019 to 2023. The figure shows 8 of the 172 counties

Tuning hyperparameters by hold-out validation

Training	Validation	Test
2019 to 2020	2021: chooses the setting	2022: scored once

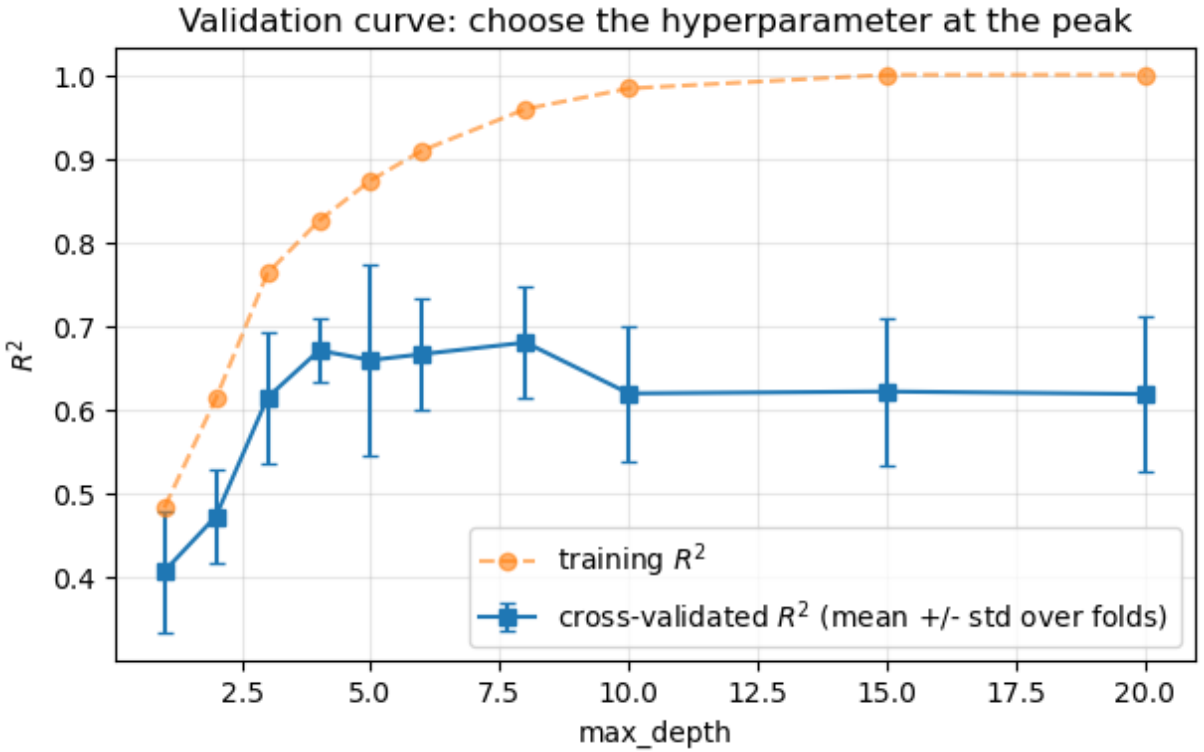




- **A hyperparameter is a setting fit does not learn:** depth, leaf size, penalty strength, number of trees
- **The rule:** the data that choose it cannot be the data that score the model. Picking the best test score picks a lucky setting
- **The three-way split:** train, validate to choose, test once at the end
- **The tree's curve bottoms out at depth 10,** so this is the optimal hyperparameter

EPA Air Quality System daily maximum 8-hour ozone, county means, 172 counties in 50 states, 2019 to 2023. Training 2019 to 2020, validation 2021, test 2022

Tuning by cross-validation: GridSearchCV




```
grid = {"max_depth": [1, 2, 3, 4, 5, 6, 8, 10, 15, 20]}
search = GridSearchCV(DecisionTreeRegressor(), grid,
                      cv=KFold(5, shuffle=True), scoring="r2")
search.fit(X_train, y_train)          # 10 depths x 5 folds
search.best_params_, search.best_score_, search.score(X_test, y_test)
```

- **What it takes:** a model, a dictionary of candidate values for its settings, a splitter, and a score
- **What it does:** for every combination in the grid, k-fold cross-validation on the training rows. Ten depths and five folds is 50 fits, and never a test row
- **What it keeps:** best_params_, the winning setting. best_score_, its mean fold score. best_estimator_, that model refit on all the training rows. cv_results_, the whole table
- **Two conventions:** scikit-learn maximizes, so errors come back negated. Several settings in one grid means every combination, so three settings with five values each is 125 candidates
- **The curve plateaus between depths 4 and 8.** The error bars say the choice inside the plateau is noise

After the search: inspect the winner, test once

Randomized search, 16 candidates	depth	leaf size	feature fraction	CV RMSE	± std
rank 1	20	8	0.58	6.11	0.30
rank 2	34	14	0.44	6.12	0.29
rank 3	16	10	0.95	6.16	0.30
rank 4	16	17	0.85	6.16	0.30
rank 16	5	18	0.91	6.84	0.31

Final forest, 2022, RMSE (ppb) by category	Forest	Persistence
West	5.13	6.31
East and Midwest	6.13	8.65
Texas and south-central	6.83	9.19
December to February	4.82	6.08
June to August	6.88	9.06
Days at or above the 70 ppb standard (376)	12.96	13.23

- **Read cv_results_, not only best_params_.** The top four candidates sit within 0.05 ppb of each other and well inside one standard deviation: a plateau, not a point
- **Inspect best_estimator_.** It is already refit on all the training data. Feature importances, and the errors by category: by region, by season, not only the mean
- **The forest beats persistence in every region and season,** but on the 376 exceedance days it is barely better and 10 ppb low on average. The score alone would not have said so
- **Then one test score, and stop.** do not tweak after measuring the generalization error, or you are overfitting the test set

Géron, Hands-On Machine Learning, 3rd ed., Ch. 2, Randomized Search, Analyzing the Best Models and Their Errors, Evaluate Your System on the Test Set. Top: forest searched with TimeSeriesSplit on 2019 to 2021. Bottom: final forest on 2022. EPA AQS ozone